A vegan or vegetarian diet substantially alters the human colonic faecal microbiota.

BACKGROUND/OBJECTIVES: Consisting of ≈10(14) microbial cells, the intestinal microbiota represents the largest and the most complex microbial community inhabiting the human body. However, the influence of regular diets on the microbiota is widely unknown. SUBJECTS/METHODS: We examined faecal samples of vegetarians (n=144), vegans (n=105) and an equal number of control subjects consuming ordinary omnivorous diet who were matched for age and gender. We used classical bacteriological isolation, identification and enumeration of the main anaerobic and aerobic bacterial genera and computed absolute and relative numbers that were compared between groups. RESULTS: Total counts of Bacteroides spp., Bifidobacterium spp., Escherichia coli and Enterobacteriaceae spp. were significantly lower (P=0.001, P=0.002, P=0.006 and P=0.008, respectively) in vegan samples than in controls, whereas others (E. coli biovars, Klebsiella spp., Enterobacter spp., other Enterobacteriaceae, Enterococcus spp., Lactobacillus spp., Citrobacter spp. and Clostridium spp.) were not. Subjects on a vegetarian diet ranked between vegans and controls. The total microbial count did not differ between the groups. In addition, subjects on a vegan or vegetarian diet showed significantly (P=0.0001) lower stool pH than did controls, and stool pH and counts of E. coli and Enterobacteriaceae were significantly correlated across all subgroups. CONCLUSIONS: Maintaining a strict vegan or vegetarian diet results in a significant shift in the microbiota while total cell numbers remain unaltered.